

Lecture 4: Functions of Random Variables

$X$ : discrete r.v. with range space (or, support)  $\mathcal{X}$

Let  $Y = g(X)$ , for a real-valued function  $g$  &  $\mathcal{Y}$  = range space of  $Y$

So,  $g: \mathcal{X} \rightarrow \mathcal{Y}$ , and  $g^{-1}(y) = \{x \in \mathcal{X} : g(x) = y\}$

In general, for  $A \subset \mathcal{Y}$ ,  $g^{-1}(A) = \{x \in \mathcal{X} : g(x) \in A\}$ .

Then ~~the~~ p.m.f. of r.v.  $Y$  from p.m.f. of  $X$  obtained as

$$P(Y=y) = \sum_{x: g(x)=y} P(X=x) = \begin{cases} \sum_{x \in g^{-1}(y)} P(X=x) & y \in \mathcal{Y} \\ 0 & y \notin \mathcal{Y} \end{cases}$$

e.g.,  $X \sim \text{Bin}(n, p)$ ,  $Y = n - X$ , then  $P(Y=y) = \sum_{x: g(x)=y} f_X(x) = f_X(n-y)$ , so that  $Y \sim \text{Bin}(n, 1-p)$

Note: We may not always get a known distribution for the transformed r.v.

e.g.,  $X \sim P(\lambda)$ ,  $Y = X^2 + 3$ ,  $\mathcal{Y} = \{3, 4, 7, \dots\}$

$X$ : continuous r.v. with  $\mathcal{X} = \{x : f_X(x) > 0\}$   
 $Y = g(X)$ , then  $\mathcal{Y} = \{y : g(x) = y, x \in \mathcal{X}\}$ .

c.d.f. of  $Y$  from c.d.f. of  $X$   
 $F_Y(y) = P(Y \leq y) = \sum_{x: g(x) \leq y} f_X(x) dx$   
 $= \begin{cases} \int_{g^{-1}(y)}^{\infty} f_X(x) dx & \text{if } g(x) \uparrow x \\ -\infty & \text{if } g(x) \downarrow x \end{cases}$   
 if  $g(x)$  is strictly monotonic, then, further

i.e.,  $F_Y(y) = \begin{cases} F_X(g^{-1}(y)) & \text{if } g(x) \uparrow x \\ 1 - F_X(g^{-1}(y)) & \text{if } g(x) \downarrow x \end{cases}$

e.g., ①  $X \sim U(0,1)$ ,  $Y = -\ln X$ , then  $F_Y(y) = 1 - F_X(e^{-y}) = 1 - e^{-y}$ ,  $y > 0$   
 $\Rightarrow Y \sim \text{Exp}(1)$ .

②  $X \sim N(0,1)$ ,  $Y = X^2$  (non-monotonic).

$F_Y(y) = P(-\sqrt{y} \leq X \leq \sqrt{y}) = 2\Phi(\sqrt{y}) - 1$ , further  
 $f_Y(y) = \frac{\phi(\sqrt{y})}{\sqrt{y}} = \frac{1}{\sqrt{2\pi}} e^{-y/2} y^{-1/2}$ , so that  $Y \sim \text{Gamma}(\frac{1}{2}, 2)$   
 or  $\chi^2_1$  distn.

Directly to pdf ~~by~~ by the Jacobian method  
(only for continuous r.v.s)

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From c.d.f. of  $Y = g(X)$ ,  $F_Y(y) = F_X(g^{-1}(y))$  if  $g(x) \uparrow x$

$1 - F_X(g^{-1}(y))$   $\downarrow$  , already seen.

$$\text{Then } f_Y(y) = \begin{cases} f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y) & \text{if } g(x) \uparrow x \\ -f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y) & \downarrow \end{cases}$$

$$\text{so that } \underline{f_Y(y)} = \begin{cases} f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right| & \forall y \in \mathcal{Y} \\ 0 & \text{on } \mathcal{Y}^c \end{cases}$$

e.g.  $X \sim U(0,1)$ ,  $Y = -2\ln X$ , obtain  $f_Y(y)$  by the Jacobian method  
 $\left| \frac{d}{dy} g^{-1}(y) \right|$  is the Jacobian of transform

### Some important distributions through transformations

1.  $X_1, \dots, X_n$  i.i.d  $N(0,1)$  variables

For any  $i$ ,  $X_i^2 \sim \chi_1^2$  (already seen)

Then  ~~$\sum_{i=1}^n X_i^2 \sim \chi_n^2$~~  (use mgf method to verify).

2.  $U \sim N(0,1)$ ,  $V \sim \chi_n^2$ ,  $U$  &  $V$  are independent, then

$$T = \frac{U}{\sqrt{V/n}} \sim t_n \quad \begin{matrix} \text{(Jacobian method)} \\ \text{to have a } 2 \times 2 \text{ Jacobian matrix} \\ \text{introduce a dummy variable.} \end{matrix} \quad \begin{matrix} (U, V) \xrightarrow{g} (T, V) \\ J = \frac{\partial(u, v)}{\partial(t, v)} \end{matrix}$$

In this case  $V$  works fine!

Ex obtain the pdf of  $T$

3.  $U \sim \chi_m^2$ ,  $V \sim \chi_n^2$  with  $U$  &  $V$  independent, then  
(can be verified as above)

$$F = \frac{U/m}{V/n} \sim F_{m,n}$$

$$\underline{\text{Note}} \quad \frac{1}{F} \sim F_{n,m}, \quad T^2 \sim F_{1,n}$$

### Application in sampling from normal distribution

$X_1, \dots, X_n$  random sample from  $N(\mu, \sigma^2)$ . Then

i)  ~~$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$~~   $\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \sim N(0,1)$  (use mgf method for  $\sum X_i$ , then  $\bar{X}$ )

ii)  $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$ ,  $\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$  (Jacobian method, but difficult. See Casella-Berger).

iii)  $\bar{X}$  &  $S^2$  are independently distributed.

Note (i) - (ii)  $\Rightarrow \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$  (Why is the loss of 1 d.f.?)

Distributions of some order statistics  
(in terms of the unordered).

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$X_1, \dots, X_n$  iid observations from a distribution with cdf  $F_X$  and pmf/pdf  $f_X$ .

~~Case~~  $X_{(1)} \leq \dots \leq X_{(n)}$  is the corresponding set of order statistics.

$X_{(r)}$  =  $r$ th smallest of  $\{X_1, \dots, X_n\}$

$$X_{(1)} = \min_{i=1, \dots, n} \{X_i\}, \quad X_{(n)} = \max_{i=1, \dots, n} \{X_i\}.$$

c.d.f. of  $Y = X_{(1)}$

$$\begin{aligned} F_Y(y) &= P(X_{(1)} \leq y) = 1 - P(X_{(1)} > y) \\ &= 1 - \prod_{i=1}^n P(X_i > y) \quad (\because \text{indep.}) \\ &= 1 - (1 - F_X(y))^n \quad (\because \text{identically dist.}) \end{aligned}$$

pmf of  $Y$ ,  $f_Y(y) = F_Y(y) - F_Y(y-)$   $\forall y \in \mathcal{Y}$

pdf of  $Y$ ,  $f_Y(y) = F'_Y(y) = n(1 - F_X(y))^{n-1} f_X(y)$ ,  $y \in \mathcal{Y}$ .

c.d.f. of  $Z = X_{(n)}$

$$\begin{aligned} F_Z(z) &= P(X_{(n)} \leq z) = \prod_{i=1}^n P(X_i \leq z) \quad (\because \text{independent}) \\ &= (F_X(z))^n \quad (\because \text{identically dist.}) \end{aligned}$$

pmf  $f_Z(z) = F_Z(z) - F_Z(z-)$

pdf  $f_Z(z) = n(F_X(z))^{n-1} f_X(z)$ ,  $z \in \mathcal{Z}$

c.d.f. of  $X_{(r)}$

$$\begin{aligned} F_{X_{(r)}}(x) &= P(X_{(r)} \leq x) = P(\text{at least } r \text{ } X_i\text{'s} \leq x) \\ &= \sum_{j=r}^n P(\text{exactly } j \text{ } X_i\text{'s} \leq x) \\ &= \sum_{j=r}^n \binom{n}{j} (F_X(x))^j (1 - F_X(x))^{n-j} \quad (\text{why??}) \\ &= I_{F_X(x)}(r, n-r+1) \quad \xrightarrow{(*)} \quad (\text{Try to verify this!}) \end{aligned}$$

(Relationship between cumulative binomial probabilities & the incomplete beta integral)

Note for  $0 < p < 1$ ,  $I_p(a, b) = \frac{1}{B(a, b)} \int_0^p u^{a-1} (1-u)^{b-1} du$

(Similar relationship between cumulative Poisson probs & incomplete gamma integral)

Using (\*) pmf of  $X_{(r)}$ ,  $f_{X_{(r)}}(x) = \frac{1}{B(r, n-r+1)} \int_0^{F_X(x)} u^{r-1} (1-u)^{n-r} du$

pdf  $f_{X_{(r)}}(x) = \frac{d}{dx} F_{X_{(r)}}(x) = \frac{1}{B(r, n-r+1)} \frac{d}{dx} \left\{ \int_0^{F_X(x)} u^{r-1} (1-u)^{n-r} du \right\}$

$= \frac{1}{B(r, n-r+1)} (F_X(x))^{r-1} (1-F_X(x))^{n-r} f_X(x)$